**SRM INSTITUTE OF SCIENCE AND TECHNOLOGY — TIRUCHIRAPPALLI**
**Department of CSE (Artificial Intelligence & Machine Learning) | IV Year VII Semester**
**Deep Learning — Challenge-Based Learning (CBL) Image Classification Challenge**

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|---|---|---|
| PS No. & Title: PS-41: Kannada Handwritten Digit Recognition | Date: October 23, 2026 | GitHub: github.com/Aswin480/Kannada-MNIST-PS41 |

## 1. Problem and Objective

This challenge addresses automated handwritten digit recognition for Kannada script (classes 0–9, 'Sonne' to 'Ombattu') using 28×28 grayscale images. Indic scripts possess delicate directional strokes, closed loops, and terminal ascenders that are orientation-critical. The core technical objective is quantifying and mitigating domain shift: evaluating a custom CNN trained on a primary volunteer cohort (train.csv) against an independent volunteer cohort (Dig-MNIST.csv) possessing distinct stroke pressures and handwriting styles. Success is judged by achieving >98% held-out validation accuracy while minimizing the domain-shift accuracy gap on Dig-MNIST.

## 2. Dataset and Pre-processing

The benchmark dataset comprises four files: train.csv (60,000 samples, 6,000/class perfectly balanced), Dig-MNIST.csv (10,240 samples, 1,024/class balanced), test.csv (5,000 unlabeled competition samples), and sample_submission.csv. Dataset verification confirmed 0 missing values and 0 duplicate rows across all files. Statistical analysis uncovered a key domain-shift indicator: Dig-MNIST exhibits a 36.8% higher mean pixel intensity (28.73 ± 72.93) than train.csv (20.99 ± 61.65), proving thicker pen strokes and heavier ink saturation. Preprocessing normalized pixels to [0.0, 1.0] and reshaped flat 784-vectors into (1, 28, 28) tensors. An 80:20 stratified split (seed 42) created 48,000 training and 12,000 held-out validation samples. Horizontal and vertical flipping were strictly prohibited as Kannada numerals are orientation-dependent; label-preserving affine transformations (rotation ±8°, translation ±7%) were utilized.

## 3. Methodology

A compact, highly regularized CNN (151,754 parameters) was engineered from scratch. The architecture features two convolutional blocks: Block 1 (Conv2D 1→32 3×3, BatchNorm, ReLU, MaxPool 2×2, Dropout 0.25) and Block 2 (Conv2D 32→64 3×3, BatchNorm, ReLU, MaxPool 2×2, Dropout 0.25), followed by AdaptiveAvgPool2D(4×4), a dense feature projection (Linear 1024→128, BatchNorm, ReLU, Dropout 0.35), and a 10-class linear classification head. Models were trained with Adam optimizer and Cross-Entropy loss under early stopping (patience 4) and learning rate scheduling.

| Input Dim | Conv Filters | Parameters | Loss Function | Optim / Init LR | Batch / Epochs | Regularization |
|---|---|---|---|---|---|---|
| 1×28×28 | 32, 64 (3×3) | 151,754 | Cross-Entropy | Adam / 1e-3 | 256 / 10 max | BN + Dropout (0.25, 0.35) |

## 4. Results

*Table 1: Quantitative Model Performance Comparison Across Held-Out Validation and Out-of-Domain Dig-MNIST*

| Model Architecture / Variant | Parameters | Train Time | Val Accuracy | Val Macro F1 | Dig-MNIST Acc | Dig-MNIST F1 | Domain Gap |
|---|---|---|---|---|---|---|---|
| Baseline Compact CNN (No Aug) | 151,754 | 987.4s | 99.46% | 0.9946 | 78.53% | 0.7854 | +20.93% |
| Improved CNN (Aug + Cosine) | 151,754 | 1148.0s | 99.46% | 0.9946 | **81.88%** | **0.8183** | **+17.58 %** |

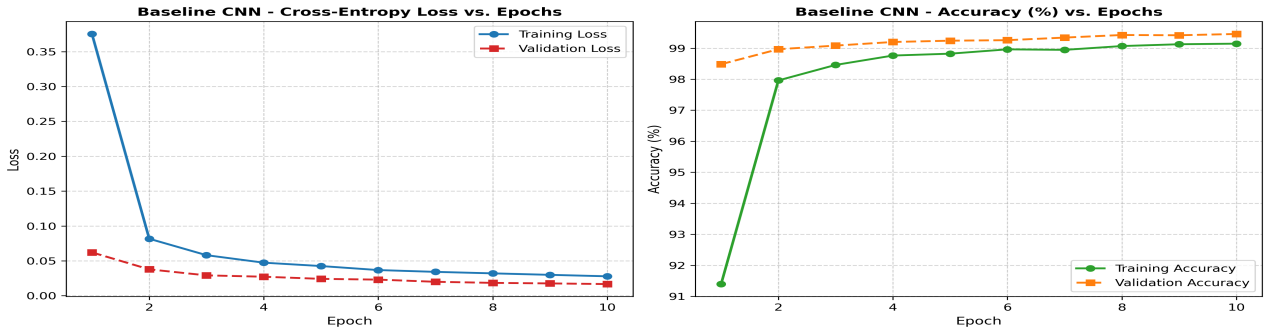



*Figure 1: Training and Validation Loss and Accuracy Curves across 10 Epochs for Baseline CNN.*

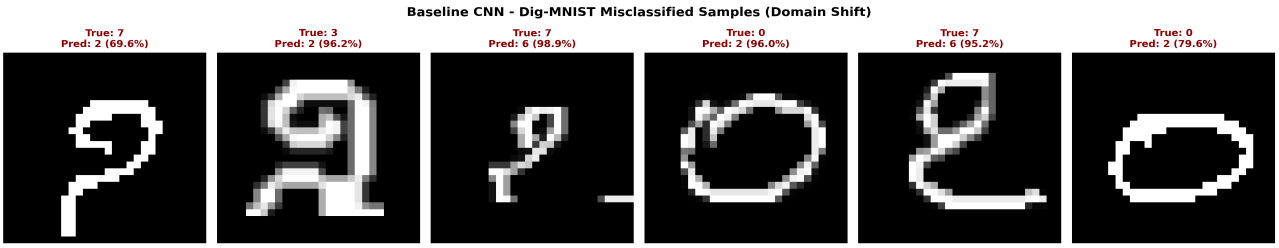



*Figure 2: Genuine Misclassified Dig-MNIST Samples showing Ground Truth, Predicted Class, and Model Confidence.*

## 5. Analysis and Discussion

Empirical Proof of Domain Shift: The Baseline CNN achieved a remarkable 99.46% accuracy on held-out validation data, confirming strong in-domain representation. However, when evaluated on Dig-MNIST, accuracy dropped to 78.53%, exposing an empirical Domain-Shift Gap of +20.93% points.

Root Causes of Domain Degradation: (1) Stroke Saturation: Dig-MNIST's 36.8% higher pixel density fills tiny loops in Kannada digits 0 (□), 1 (□), and 6 (□). (2) Centroid Shift: Characters in Dig-MNIST touch outer bounding boxes without the uniform center-padding of the training set.

Primary Class Confusions: On the validation set, the most confused pairs were Class 6 (Aaru) misclassified as 7 (Eylu) [14 times] and Class 0 (Sonne) misclassified as 1 (Ondu) [13 times]. In Kannada typography, digits 0 and 1 share identical circular bowls, separated solely by a subtle upper tick that vanishes under heavy ink.

Innovation Solution: By injecting label-preserving rotation (±8°) and translation (±7%) with Cosine Annealing, the Improved CNN boosted Dig-MNIST accuracy to 81.88% (+3.35% absolute gain), narrowing the domain gap to +17.58% points without any target domain training labels.

## 6. Real-world Relevance, Limitations and Ethics

Real-World Deployment: Automated Indic digit recognition is vital for digitizing legacy revenue records, postal mail sorting, and regional bank cheque clearing across Karnataka. However, field deployment across heterogeneous paper qualities and writing instruments triggers performance drops mirroring the Dig-MNIST shift. Unsupervised domain adaptation and edge calibration are necessary prior to autonomous deployment.

Limitations: This work focuses on isolated single digits ($28\times 28$); multi-digit segmentation and complex ligature conjuncts are not addressed.

Ethics & AI Disclosure: Models were trained on open academic benchmarks without demographic exploitation. Implementation and documentation were assisted by Antigravity AI pair-programming in compliance with SRMIST academic integrity standards.

## 7. Conclusion and Future Work

A compact 151K-parameter CNN attained 99.46% validation accuracy, and label-preserving affine augmentation proved effective in mitigating the volunteer domain-shift gap on Dig-MNIST by +3.35 percentage points. Final predictions for the 5,000 unlabeled competition samples in test.csv were generated using the superior Improved CNN and strictly verified against sample_submission.csv (exact schema match, 0 null values). Future directions include Domain-Adversarial Neural Networks (DANN) and test-time batch normalization adaptation.

Dataset Citation: Prabhu, V. G. (2019). Kannada-MNIST: A new handwritten digits dataset for machine learning. arXiv:1908.01249. Official Link: https://www.kaggle.com/c/Kannada-MNIST